Section A

Cluster 2 Fieldwork

1 Green Hub @ Bishan is a community project located at the void deck of a HDB block in Bishan to promote sustainability among its residents. It houses an urban community garden and a 24-hour community fridge to encourage food rescue efforts.

A group of 4 Geography students from Singapore decided to investigate the extent to which Green Hub @ Bishan has encouraged sustainable practices among the residents living in its vicinity. They developed the following hypothesis for their fieldwork:

"Green Hub @ Bishan is effective in encouraging sustainable practices among residents living in its vicinity."

The students visited Green Hub @ Bishan on a Friday afternoon from 3pm to 5pm. They decided to split themselves into two groups to conduct their fieldwork.

1 student stayed at the Green Hub and did a count of the number of residents who either contributed to the community fridge or took food from the community fridge for the duration of their fieldwork investigation. Over 2 hours, the student counted 12 residents in total who either contributed to or took food from the community fridge.

The remaining students conducted questionnaire surveys with residents in the vicinity of the Green Hub to find out if the Green Hub had encouraged them to adopt sustainable practices. They split themselves up between Block 112 (where the Green Hub is located) and Block 113. They used convenience sampling and surveyed 42 residents in total. When they analysed their survey findings, the students noted that 25 residents were aged above 60.

Resource 1 shows the map of the neighbourhood estate where Green Hub @ Bishan is located and the features of Green Hub @ Bishan. Resource 2 shows the recording sheet for the survey. Resource 3 is an extract of the data collected from the survey.

(a) Suggest **one** limitation of the hypothesis for the investigation. [2]

Possible pointers:

- The hypothesis lacks specificity in defining what is under investigation. [1 mark]
  - What constitutes a "sustainable practice" is not defined and up to interpretation. [1 additional mark]
  - What constitutes the "vicinity of Green Hub @ Bishan" is not defined and up to interpretation. [1 additional mark]

- The hypothesis might not be at a suitable scale for the investigation. [1 mark]
  - Given that there are only 4 students in the team investigating within 2 hours, it may be too large an area for 4 students to cover the entire vicinity of Green Hub @ Bishan. [1 additional mark]
  - The investigation was only conducted over a period of 2 hours from 3pm – 5pm, which is too short to cover all the areas under investigation in the vicinity of Green Hub @ Bishan [1 additional mark]

(b) Explain how the students could minimise the risks associated with their fieldwork. [4]

Possible pointers:

- At least 1 student conducted the fieldwork investigation alone. [1 mark] To minimise the risk of incidents happening without quick access to help (e.g. disputes or harassment from people passing by), the students should conduct their fieldwork in pairs. [1 mark]
- Based on Resource 1A, there are many small roads in the vicinity of Green Hub @ Bishan to get to the other blocks. [1 mark] To minimise the risk of small road accidents occurring, students should be on the lookout for vehicles every time they cross the road and to alert their groupmates about incoming vehicles. [1 mark]

(c) Suggest why the sampling conducted for the survey was not representative of the residents living in the vicinity of Green Hub @ Bishan. [4]

Possible pointers:

- *[Lack representativeness in sample size]* The number of residents might have been too small relative to the number of residents actually living in the HDB blocks in the vicinity of Green Hub @ Bishan [1 mark]. This can be seen from the fact that only 42 residents were surveyed, which is too few considering the number of residents living in a typical HDB block. [1 additional mark]

- *[Lack representativeness in age profile]* The proportion of people surveyed by age group might not have been representative of the age profile of the residents living in the vicinity of Green Hub @ Bishan. [1 mark] The majority of the respondents (25 out of the 42 residents surveyed or nearly 60% of respondents) were aged above 60 years old, which might have caused the views of this age group to be overrepresented in the survey findings. [1 additional mark]

- *[Lack representativeness in spatial scope]* The blocks at which residents were surveyed were too few relative to the number of HDB blocks found in the vicinity of Green Hub @ Bishan and may hence not be representative of the area under investigation. [1 mark] Only residents who passed by Blocks 112 and 113 were surveyed, but from Resource 1A, there are many other blocks in the direct vicinity of Green Hub @ Bishan. [1 additional mark]

(d) With reference to Resource 2, explain the strengths and limitations of the recording sheet used for the survey**.** [5]

Possible pointers:

Strengths	Limitations
• *[Number and length of questions]* Both the number of questions (3 questions only) and the length of questions are short (Q1 and Q2 are short statements) [1 mark], which would encourage more respondents to respond to the survey instead of abandoning it due to lack of time or attention to sustain the survey. [1 mark]	• *[Ambiguous options]* Some options for response in the recording sheet are vague and subject to differences in interpretation – e.g. what is meant by “often” or “sometimes” for Question 2 would differ from respondent to respondent [1 mark]. This would reduce the accuracy of the findings from the survey [1 mark].

• *[Capturing profile of residents]* The recording sheet allows data to be collected for gender, age and occupation [1 mark], allowing the data collected to be analysed across different social profiles for more detailed analysis. [1 mark]	• [All closed questions] All the questions in the survey are closed questions. [1 mark] This would mean that respondents do not have any avenue to contextualise their responses and provide further information that might be helpful in the interpretation of the results [1 mark].

Award max. of 3 marks if **only** strengths or limitations was discussed.

(e) Explain one limitation of the data representation method seen in Resource 3 and sketch a diagram to better represent the data collected for Question 3**.** [5]

Possible pointers:

- Limitation:
  - The table does not allow for **<u>easy visual comparison</u>** of the results for analysis. [1 mark]
- Sketch diagram:
  - Choice of suitable data representation method (e.g. pie chart) [1 mark]
  - Accurate labels [1 mark]
  - Accurate values / proportions [2 marks – 1 mark per 2 accurate portions]

(f) The students concluded that Green Hub @ Bishan has been largely effective in encouraging sustainable practices for the residents living in its vicinity.

Evaluate the validity of their conclusion. [10]

Possible pointers:

Conclusion could be argued to be valid because…	• Hypothesis aligns with the overall aim of the fieldwork investigation, which provides direction for the investigation, ensuring that the conclusion is relevant to the aim of the fieldwork • Conclusion does somewhat align with the results seen in Resource 3 ○ E.g. Majority of respondents (24 out of 42) agreed that Green Hub @ Bishan had encouraged them to adopt sustainable practices.
Conclusion could be argued to be invalid because…	• Sampling was not representative of the profile of the residents. As fieldwork only held on one weekday afternoon for 2 hours, many of the residents were disproportionately of the older age group, 60 years and above, as working adults and students are likely not to be at home. This reduces the reliability of the findings

	because the findings are skewed towards the perceptions of elderly residents. • Scope of fieldwork was limited, as the investigation only covered 2 blocks, though there are many other blocks in the direct vicinity of Green Hub @ Bishan, reducing the reliability of the findings. • Inaccurate data collected due to design of recording sheet, as some of the options featured ambiguous wording that is subject to interpretation. • Some of the findings are inconclusive OR contradict the conclusion: o Inconclusive – E.g. 12 people were seen to either take food from or contribute to the community fridge, but the significance of this number is unclear (Preamble) o Contradicts conclusion – E.g. 15/ 42 of the respondents surveyed (more than half) did not use the community fridge (Based on Question 2 of Resource 3)

Section B

Cluster 1 Sustainable Future and Climate Change

2 Resource 4 shows population growth in urban and rural areas in more developed regions and less developed regions from 1950 to 2030.

Cañada Real is a 15-km informal settlement located in Madrid, Spain. With over 8000 inhabitants most dwellers live below the poverty line in informal, low-quality housing. Resource 5 shows an aerial view of the Cañada Real settlement and an example of a dwelling in the settlement. Resource 6 provides information on the living conditions in the Cañada Real settlement.

(a) With reference to Resource 4, compare the population growth trends between more developed and less developed regions from 1950 t0 2030. [5]

Things to note:

- *At least 2 bases of comparisons should be identified.*

Possible points:

More developed	Basis of comparison	Less developed
From about 500 million in 1950 to 1000 million in 2030.	**Similarity: Both have experienced an increase in urban areas.** **[1m]**	From about 400 milion in 1950 to 3900 million in 2030.
Slower rate of increase - Increase by about 2 times (500 million) over the 80 years period.	**Difference: Rate of increase in urban areas differ.** **[2m]**	**Faster rate of increase**: Increased almost 10 times (3500 million) over the 80 years period.
Decline: From about 400 million in 1950 to 250 million in 2030.	**Difference: Nature of change in rural areas differ.** **[2m]**	**Mostly increase, with a slight decline from 2015:** • Increase from about 1400 million in 1950 to about 3100 million in 2015, before declining to about 3000 million in 2030.

(b) Explain **one** implication of urban population loss on progress towards sustainable urban development. [3]

Things to note:

- *Award 1m for identification of 1 dimension of SUD that might be affected.*
- *Award 2m for explanation + further development of explanation.*

Possible points:

- **Impact on environmental sustainability** – With a decline in population, there is an increased probability of property abandonment which suggests that the built environment is likely to experience neglect/ will not be maintained, and services such as waste disposal might also cease, causing environmental degradation.

- **Impact on social sustainability** – Urban population loss is likely to result in ageing of urban areas as the younger population are more willing/able to move out of the city areas. This might result in greater demand for healthcare and other social support services by the elderly which might not be able to be met adequately if city authorities lack the financial means/ willingness to do so.

- **Impact on economic sustainability** – A decline in population suggests a loss of potential labour as well as consumers, hence, this might result in a movement out of existing business and deter new investments into the area. This will worsen unemployment as well, reducing expenditure by the locals, and perpetuating the probability for more businesses to leave.

(c) With reference to Resource 5, describe the characteristics of the Cañada Real informal settlement. [4]

Things to note:

- *Award 1 mark for each description of the characteristics, to a maximum of 4 marks.*
- *Award a maximum of 1 additional mark for further development of each description, where applicable.*

Possible points:

- **From the aerial photograph (max 2m):**
    - Linear development of infrastructure, with rather haphazard organisation (1m)
    - Moderate density (units look fairly close to each other) (1m).
    - There is also some evidence of accumulation of rubble/waste on the slopes running parallel to the settlement in the right middle-ground (1m), suggesting a lack of waste collection services. (1 additional mark).

- **From the example of the individual dwelling (max 2m):**
    - Makeshift shelter comprising what seems to be unwanted/ discarded materials (1m) such as canvas/plastic sheets on top of the dwelling and serving as a makeshift door as seen in the centre to right middle-ground (1 additional mark).
    - There is also evidence on rubbish strewn around the unpaved path in front of the dwelling. (1m)

(d) With reference to Resources 5 and 6, explain how the deprivations experienced by residents in the Cañada Real settlement can impact sustainable urban development. [6]

Things to note:

- *Award 1 mark for each suggestion, with reference to Resources 5 & 6, on how deprivations may impact SUD.*
- *Award a maximum of 1 additional mark for further development of each suggestion.*

Possible points:

From Resource 5 [max 4m]

- Poor standard of housing and living environment - the cramped and overcrowded conditions as well as the unsafe and/or weak housing structures → affects the social dimension of SUD (1m).
    - Cramped and overcrowded housing conditions can easily cause and exacerbate the transmission of diseases (1m)
    - The lack of privacy among dwellers of the same housing unit could negatively impact some slum dwellers' mental health (1m).
    - Unsafe and weak housing structures too can be physically hazardous to slum dwellers, especially for those built in environments that are susceptible to natural hazards such as landslides and earthquakes (1m).

From Resource 6 [max 4m]

- Majority have irregular connection to electricity supply (93%), and lack access to street lighting (95%) → affects the social/economic dimensions of SUD (1m).
    - *Social dimension*: Affects daily activities at night, and could compromise the personal safety of residents (e.g. increased risk of accidents/hazards when moving about the settlement; increased risk of harassment for female residents, etc.). (1m)
    - *Social & economic dimension*: Hinders the learning of school-going of children/youth who may not be able to engage in learning activities at night, affecting their attainment of higher educational levels (1m) → in the long-term, could perpetuate the cycle of poverty since the lower educational attainment may limit them to less well-paying jobs (1 additional mark).
    - *Economic dimension*: Limits economic activity to daylight hours or lowers the productivity of any economic activity that requires the use of electrical machinery → limits earnings. (1m)

- Majority (70.4%) do not have access to sanitary sewer systems which are more hygienic sanitation disposal systems → affects the social dimensions of SUD as it increases potential health risks (1m).
    - Most households (44.5%) use a cesspool for their sanitary waste, and this leads to contamination of the environment since cesspools have an open bottom/ perforated sides (1m) → increases the chance of transmission of diseases or contamination of water supplies, both of which jeopardises the health of residents (1 additional mark).
    - Slightly more than a quarter (25.9%) of households are without sanitation, and this suggests that they are likely to have to resort open/exposed areas for toileting needs (1m) → this increases the chances of health risks due to the accumulation of waste (1 additional mark) as well as increased risks to safety, especially for children/women who may be subject to harassment (1 additional mark).

(e) Explain why slums may develop in urban areas of less developed countries**. [AO1]** [4]

Things to note:

- *Award 1 mark for each explanation of why urban areas develop in urban areas of less developed countries.*
- *Award a maximum of 1 additional mark for further development of each explanation.*

Possible points:

- **High rates of urban population growth & role of migration**
  - Demand for housing often far outweighs the available supply of formal housing units as high rates of RUM persist due to significant push-pull factors that exist between rural and urban areas → e.g. job prospects; provision of social amenities etc
- **Urban poverty and lack of affordability of housing**
  - Decrease in formal employment opportunities
  - Higher cost of living in urban centres
  - Commercialisation/ privatisation of goods and services
- **Failure of governance to provide affordable housing**
  - Inability to respond to rapid urbanization which occurs at a pace that far exceeds the planning process to incorporate new migrants into the city
  - General inertia on the part of urban authorities in taking steps to tackle the illegal occupation of urban lands, often due to the scale, sensitivity, and/or complexity of the issue → their sheer magnitude overwhelms the capacity/capability of existing programmes or policies to effectively address the issue
  - Lack of acknowledgement of rights of urban poor → Governments often fail to incorporate them into urban planning, thereby contributing to the growth of slums.

3 Resource 7 shows the Gender Equality score by region, from 1975 – 2020. Resource 8 shows the hours spent on leisure, housework, and work by females and males in Japan, segmented by age group. Resource 9 shows the percentage difference in full-time earnings between men and women in selected OECD nations in 2016. Countries in the Organisation for Economic Co-operation and Development (OECD) are generally considered to be developed countries. Resource 10 depicts a strategy implemented in Tokyo, Japan to address issues faced by women living in the city.

(a) Explain the difference between sustainable urban development and urban liveability. [4]

Award 2 marks for every identification of difference + explanation of difference.

Possible pointers:

Sustainability	Basis of Comparison	Liveability
Application on **a global geographic scale** (i.e. on a macro, big picture scale)	**Spatial scale**	Grew out of a view that environmental, economic, and equity issues must be considered at a narrower spatial scale relevant to individual people, households, neighbourhoods, and communities in geographically smaller areas, i.e. **local environments**
Long-term and intergenerational time horizons	**Time**	Immediate and tangible conditions and interventions
Sustainable development involves an **unchanging** vision that joins economic, environmental, and equity values → in other words, it provides a long-term goal/ vision.	**Context of application**	Urban liveability is **dynamic** and evolves in response to shifting conditions and values → in other words, it facilitates implementation of specific interventions.

(b) Using Resource 7, compare the Gender Equality score between the North America region and the Asia/Pacific region from 1975 – 2020. [4]

Possible pointers:

North America	Basis of Comparison	Asia/Pacific region
Both saw an increase in the Gender Equality score. North America's score increased from 0.55 to 0.78.	Nature of change over time	Both saw an increase in the Gender Equality score. Asia/Pacific's score increased from 0.38 to 0.53.
North America's Gender Equality score saw a	Degree of change	Asia/Pacific's Gender Equality score saw a

higher degree of increase of 0.23, from 0.55 to 0.78.		lower degree of increase of 0.51, from 0.38 – 0.53.

(c) Explain the importance of gender equality for cities to become more liveable. [4]

Possible pointers:

Definition of gender equality: Gender equality refers to the equality between women and men. It refers to the equal rights, responsibilities and opportunities of women and men and girls and boys.

Gender equality is important for cities to become more liveable because:

1. Problems of gender inequality faced in cities diminishes liveability of cities
    - In order for cities to become more liveable for all its residents, not just some, gender equality in cities is necessary. [1 mark]
        - However, gender inequality remains an issue for many cities. Women and girls make up half of the population of city resident, but they continue to suffer systematic gender discrimination in cities. [1 mark]
        - Also, cities are not designed in a way that helps women] Men and women use cityscapes in different ways, leading to different experiences and perceptions of cities by both groups. [1 mark] However, more often than not, urban planning takes greater consideration of the needs of men and how they move around and experience the city as compared to women. [1 mark]
            - [Provide example – 1 additional mark]
            - E.g. Transport patterns: In the mid-1990s in Vienna, urban planners created a questionnaire that asked residents why and how they used public transportation. The results showed that men's typical route was short and simple: often to and from work, but women's responses, however, were more complex and varied, usually including multiple trips a day on public transport and as well as on foot ("multitasking") for multiple purposes – e.g. dropping off children at school, going to work, leaving work, getting groceries, back to school for pick up, etc.

2. Gender equality improves liveability of cities, as the move towards gender equality in cities will also help to improve economic wellbeing and quality of life of all residents. [1 mark]
    - For example, a study by ActionAid (2015) showed that if women in developing countries were paid as much as men and had equal access to the labour market, these countries would make an additional $9 trillion per year. [1 additional mark]
    - For example, when cities are more inclusive towards women and takes into consideration the needs of women in urban planning, this would encourage a greater number of women to enter the workforce, increasing purchasing power of the residents. [1 additional mark]

(d) With reference to Resource 8 and Resource 9, explain why gender inequality impacts the well-being of women aged 20 to 50 in Japan. [6]

Possible pointers:

From Resource 8

- Gender equality is an issue as many females in Japan bear the double burden of both domestic responsibilities and work [1 mark].
- Based on Resource 8, for this age group, women spend more hours on housework compared to men. Women spend between 3 hours to 6 hours on housework as compared to their male counterparts who spend 0.5 – 1 hour on housework each day. [1 mark]
- They might experience great stress for needing to balance multiple responsibilities at the same time, thus impacting their psychological well-being. [1 additional mark]

From Resource 9

- Gender equality is an issue as many females are paid less than their male counterparts / experience a gender pay gap in Japan. [1 mark]
- From Resource 9, there is a 26.6% difference between the full-time pay of males and females in Japan, which is the second-highest difference among selected OECD countries. [1 mark]
- As they are consistently paid less than their male counterparts, females in Japan are more likely to struggle to save money for retirement / old age, resulting in an adverse impact on their economic wellbeing in the future [1 additional mark]

(e) With reference to Resource 10, suggest how the strategy implemented addresses issues faced by women living in Tokyo, Japan. [4]

Possible pointers:

- The strategy seen in Resource 4 is that of gender-segregated cabins on public transport [1 mark].

- This strategy significantly reduces the likelihood of physical and sexual harassment by men on crowded trains given that men are not allowed in these cabins during peak hours [1 mark], thus helping women to feel safer when taking public transport, thus improving their psychological wellbeing [1 additional mark].

- This might also encourage more women to be employed or continue to be employed as they do not need to fear being on public transport while travelling to and from work [1 mark], which thus provides women the opportunity to earn an income and address issues of economic wellbeing faced by women in Tokyo [1 additional mark].

- This also helps to improve confidence of women in going out or staying out late (either to work or other social engagements) [1 mark], thus improving self-confidence of women and therefore their social well-being [1 additional mark].

Section C

Answer **two** questions

4 Evaluate the success of strategies to manage non-hazardous solid waste in urban areas of countries at different levels of development. [13]

Possible Approaches:

Candidates could approach the question by making a judgement on whether some strategies have been more effective than others in managing non-hazardous solid waste in urban areas through a consideration of criteria such as:

- The ability to address the root causes of the nature/scale of the waste issue – e.g. through ensuring a more circular urban metabolism
- The ability to bring about long-term resolution of the waste issue – e.g. by ensuring that the strategies undertaken do not generate trade-offs that compromise the sustainable urban development of the city.
- The ability to adequately address the extent/magnitude of the waste issue

Candidates could approach the question by making a judgment on whether strategies used to manage non-hazardous solid waste in urban areas have been successful through a consideration of two or more case studies.

- Candidates could evaluate whether the strategies in each case study achieved the intended aims or targets.
- Candidates could also analyse the contextual factors such as the level of development of the country and the affordability of these strategies which may have contributed to the extent of success of the strategies. Candidates could also analyse the challenges to success which are common across the case studies.

Levels marked using Generic Level Descriptors for 13m H1 essays

5 Evaluate the view that it is impossible to address the issues faced by the elderly living in cities. [13]

Possible Approaches:

Candidates could approach the question by making a judgement on whether some strategies have been effective in addressing the issues faced by the elderly living in cities through a consideration of criteria such as:

- The ability to address the root issues that elderly face or the ability to comprehensively address both the physical and psychological needs of elderly → i.e. improving liveability of the city for the elderly
- The ability to comprehensively address the issues faced by different segments/profiles of elderly
- The ability to address this issues in a sustained manner (i.e. long-term)

Candidates could approach the question by making a judgment on whether strategies used to address the issues faced by the elderly living in cities have been successful through a consideration of two or more case studies.

- Candidates could evaluate whether the strategies in each case study achieved the intended aims or targets.
- Candidates could also analyse the contextual factors such as the level of development of the country and the affordability of these strategies which may have contributed to the extent of success of the strategies. Candidates could also analyse the challenges to success which are common across the case studies.

Levels marked using Generic Level Descriptors for 13m H1 essays

6 "Mitigation is the best response to climate change and its effects."

Evaluate this statement. [13]

Possible Approaches:

- Candidates could approach the question by making a judgement on whether mitigation, as compared to adaptation, is the best response to climate change and its effects through a consideration of criteria such as:
    - Addressing the root cause of climate change
    - Spatial scale at which climate change effects are addressed (e.g. is the response applicable globally, regionally and/or nationally? is the response able to address locally-experienced effects of climate change?)
    - Temporal scale at which climate change effects are addressed (e.g. is it a long-lasting response to climate change? does it immediately address climate change effects experienced by a particular group of people?)
    - Feasibility of strategy (e.g. is it financially feasible for countries to implement? are the trade-offs when the strategy is implemented acceptable?)

- Consideration of what the "best" response should also take into account the context of the country – e.g. level of development, exposure to climate change effects, etc. – with the understanding that the idea of "best" could differ based on context

Levels marked using Generic Level Descriptors for H1 essays.